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MethodsX

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About the journal

Aims and scope

An open access journal publishing methods and protocols.

If this journal is a good fit for your methods or protocols, you can find out more via the [Guide for Authors](#) and are invited to submit your manuscript using MethodsX's [methods](#), [protocol](#) or [review](#) template.

MethodsX is a multidisciplinary, open access, peer-reviewed journal, which publishes detailed but digestible articles that describe methodological advances. It contributes to open science and improves reproducibility by making methods, protocols, reviews and the associated research more discoverable; opening doors for collaboration; and sparking new lines of inquiry to enhance the research cycle.

In support of the FAIR Data Principles, which state that research data should be Findable, Accessible, Interoperable and Reusable, *MethodsX* aims to make your methods and protocols:

- Discoverable – We will make your article easy to find and download, giving readers easy access to both your method or protocol and your original research article.
- Citable – Receive recognition for the work you've done with a peer-reviewed article with a distinct DOI. Drive traffic to and from your *MethodsX* article and original research with linking and references.

- **Reproducible and reusable** – Promote reproducibility by openly sharing methods or protocols that strengthen the reliability, transparency, validity, and rigor of results and help your peers save valuable time, resources, and funding.
- **Trustworthy** – Earn the respect of your peers with method articles reviewed by experts.

The journal welcomes high-quality submissions describing methodological advances from all subject areas, including customizations to established methods and protocols. Articles should be written using one of the provided templates and include evidence of the efficiency of the method or protocol and/or comparison with pre-existing practices. Submissions undergo thorough peer-review to check for suitability and quality.

Article types

MethodsX welcomes four types of articles: Method articles, Protocol articles, Review articles and Letters to the Editor.

- **Method articles** describe new methods in all research areas.
- **Protocol articles** focus on the life and health science areas.
- **Review articles** review and discuss methodological information.
- **Letters to the Editor** are commentaries on methods/protocols previously published.

Please select and download the correct **template** to prepare your article: [Method article template](#), [Protocol article template](#) or [Review article template](#).

A *MethodsX* article features:

- An **abstract** to outline the customization of the method or protocol.
- A **graphical abstract** to illustrate what you've done.
- The **method(s)** or **protocol(s)** in sufficient detail to help people replicate it, including any relevant figures and tables.
- Up to **25 references** to the original description of the method you're using.

You can find some examples [here](#) and our five-minute overview highlighting the most important information for authors [here](#).

Method articles

A method article is a short, templated article, describing methodological advances from all subject areas. The article can also describe customizations to previously established methods. Articles should also include evidence of the efficiency of the method. All manuscripts undergo thorough peer-review to check for suitability and quality.

Protocol articles

A protocol article is a short, templated article, describing new protocols or customizations in the fields of medicine and life sciences. Articles should also include evidence of the efficiency of the protocol. All manuscripts undergo thorough peer-review to check for suitability and quality.

Review articles

Via this article type, we invite reviews assessing current knowledge of methods used in a specific field. The review must evaluate existing methods, discuss limitations and/or propose modifications. All manuscripts undergo thorough peer-review to check for suitability and quality.

Letters to the Editor

With this article type, we invite objective and constructive commentaries on methods/protocols previously published in *MethodsX*. By publishing comments on published articles, we hope to increase the reusability of methods/protocols. Upon publication of the Letter to the Editor, the authors of the original article will be asked to write a rebuttal to the comments raised in the Letter to the Editor, and they have the possibility to update their method/protocol accordingly. Ultimately, the original method/protocol article, the Letter to the Editor and the rebuttal will all be linked on ScienceDirect. Letters to the Editor and the rebuttal are free of charge. We welcome submissions from all topics and subject areas, as we believe that Letters to the Editor benefit all types of articles.

Letters to the Editor are **not templated** but should contain comments on the previously published method/protocol, recommendations for improvement, as well as a declaration of competing interests, an ethics statement, and a CRediT author statement.

Peer review

This journal follows a single anonymized review process. Your submission will initially be assessed by our editors to determine suitability for publication in this journal. If your submission is deemed suitable, it will typically be sent to a minimum of two reviewers for an independent expert assessment of the scientific quality. The decision as to whether your article is accepted or rejected will be made by our editors.

Read more about [peer review](#).

Our editors are not involved in making decisions about papers which:

- they have written themselves.
- have been written by family members or colleagues.
- relate to products or services in which they have an interest.

Any such submissions will be subject to the journal's usual procedures and peer review will be handled independently of the editor involved and their research group. Read more about [editor duties](#).

Authors may submit a formal appeal request to the editorial decision, provided it meets all the requirements and follows the procedure outlined in [Elsevier's Appeal Policy](#). Only one appeal per submission will be considered and the appeal decision will be final.

Special issues and article collections

The peer review process for special issues and article collections follows the same process as outlined above for regular submissions, except, a guest editor may send the submissions out to the reviewers and may recommend a decision to the journal editor. The journal editor oversees the peer review process of all special issues and article collections to ensure the high standards of publishing ethics and responsiveness are respected and is responsible for the final decision regarding acceptance or rejection of articles.

Open access

We refer you to our [open access information page](#) to learn about open access options for this journal.

Ethics and policies

Ethics in publishing

Authors must follow ethical guidelines stated in [Elsevier's Publishing Ethics Policy](#).

Submission declaration

When authors submit an article to an Elsevier journal it is implied that:

- the work described has not been published previously except in the form of a preprint, an abstract, a published lecture, academic thesis or registered report. See our policy on [multiple, redundant or concurrent publication](#).
- the article is not under consideration for publication elsewhere.
- the article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.
- if accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright-holder.

To verify compliance with our journal publishing policies, we may check your manuscript with our screening tools.

Authorship

All authors should have made substantial contributions to all of the following:

1. The conception and design of the study, or acquisition of data, or analysis and interpretation of data
2. Drafting the article or revising it critically for important intellectual content.
3. Final approval of the version to be submitted.

All authors are accountable for all aspects of the work, to ensure that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Authors should appoint a single corresponding author to communicate with the journal during the editorial process. This individual submits the manuscript and all required documentation to the journal, and serves as the primary

contact between the journal and the co-authors. Only the corresponding author's affiliation will be used to determine eligibility for a publishing agreement, and possible discounts related to it. Affiliations of other co-authors are not relevant for eligibility.

Changes to authorship

The editors of this journal generally will not consider changes to authorship once a manuscript has been submitted. It is important that authors carefully consider the authorship list and order of authors and provide a definitive author list at submission.

The policy of this journal around authorship changes:

- All authors must be listed in the manuscript and their details entered into the submission system. Changes can only be made prior to acceptance, and only if approved by the journal editor. This includes additions, deletion, or rearrangement of author names.
- Requests to change authorship must be made by the corresponding author using the [Authorship Change Request form](#). The corresponding author must provide the reason for the request to the journal editor with written confirmation from all authors, including any authors being added or removed, that they agree with the changes. Requests which do not comply with the instructions outlined in the form will not be considered.
- This journal does not allow authorship changes after acceptance. This includes additions, deletions, or the rearrangement of author names, including changes to the corresponding author.
- The review process may be paused while a change in authorship request is being considered.
- Change requests made by the authors without the [Authorship Change Request form](#) or editor approval may result in the rejection of the manuscript, or retraction if the article has already been published.

Declaration of competing interests

All authors must disclose any financial and personal relationships with other people or organizations that could inappropriately influence or bias their work. Examples of potential competing interests include:

- Employment
- Consultancies
- Stock ownership
- Honoraria
- Paid expert testimony
- Patent applications or registrations
- Grants or any other funding

- Affiliation with the journal as an Editor or Advisory Board Member

The [declarations tool](#) should always be completed.

Authors with a journal affiliation to declare should enter the following text under “Other Activities” within the [declarations tool](#) and should inform the journal and publisher prior to completing the submission process:

Given their role as [AUTHOR JOURNAL ROLE TITLE], [AUTHOR NAME] had no involvement in the peer-review of this article and has no access to information regarding its peer-review. Full responsibility for the editorial process for this article was delegated to another journal editor.

Editorial disclosure statements will be included as a footnote and/or in the declaration of competing interest section of the article.

Authors with no competing interests to declare should select the option "I have nothing to declare".

The resulting Word document containing your declaration should be uploaded at the "attach/upload files" step in the submission process. It is important that the Word document is saved in the .doc/.docx file format. Author signatures are not required.

Funding sources

Authors must disclose any funding sources who provided financial support for the conduct of the research and/or preparation of the article. The role of sponsors, if any, should be declared in relation to the study design, collection, analysis and interpretation of data, writing of the report and decision to submit the article for publication. If funding sources had no such involvement this should be stated in your submission.

List funding sources in this standard way to facilitate compliance to funder's requirements:

Funding: This work was supported by the National Institutes of Health [grant numbers xxxx, yyyy]; the Bill & Melinda Gates Foundation, Seattle, WA [grant number zzzz]; and the United States Institutes of Peace [grant number aaaa].

It is not necessary to include detailed descriptions on the program or type of grants, scholarships and awards. When funding is from a block grant or other resources available to a university, college, or other research institution, submit the name of the institute or organization that provided the funding.

If no funding has been provided for the research, it is recommended to include the following sentence:

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI use

Authors must declare the use of generative AI tools in the manuscript preparation process upon submission of the paper.

Generative AI and AI-assisted technologies (“AI tools”) have been rapidly adopted in research and scholarly publishing, offering significant benefits while requiring clear standards for responsible and transparent use.

Authors preparing a manuscript for this journal may use AI tools to support them. However, these tools must never be used as a substitute for human critical thinking, expertise and evaluation. AI tools may only be applied with human oversight and control.

Please read Elsevier's detailed author policy on the use of AI tools, as stated in [Elsevier's generative AI policies for journals](#).

Ultimately, authors are responsible and accountable for the contents of their work. This includes accountability for:

- Carefully reviewing and verifying the accuracy, comprehensiveness, and impartiality of all AI-generated output (including checking the sources, as AI-generated references can be incorrect or fabricated).
- Editing and adapting all material thoroughly to ensure the manuscript represents the author's authentic and original contribution and reflects their own analysis, interpretation, insights and ideas.
- Ensuring accuracy and originality of any AI generated images and provide appropriate attribution where relevant.
- Ensuring the use of any tools or sources, AI-based or otherwise, is made clear and transparent to readers. If AI tools have been used, we require a disclosure statement upon submission; please see example below.
- Ensuring the manuscript is developed in a way that safeguards data privacy, intellectual property and other rights, by checking the terms and conditions of any AI tool that is used.

Finally, authors must not list or cite AI tools as an author or co-author on the manuscript since authorship implies responsibilities and tasks that can only be attributed to, and performed by, humans.

The use of AI tools in the manuscript preparation process must be declared by adding a statement at the end of the manuscript when the paper is first submitted. The statement will appear in the published work and should be placed in a new section before the references list. An example:

Title of new section: *Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.*

Statement: *During the preparation of this work the author(s) used [NAME OF TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.*

The declaration does not apply to the use of basic tools, such as tools used to check grammar, spelling and references. In addition, authors are not required to disclose the use of AI tools or features within specialist disability-related assistive technology, provided these are used solely for accessibility purposes.

If you have nothing to disclose, you do not need to add a statement.

Please note: to protect authors' rights and the confidentiality of their research, this journal does not currently allow academic editors and reviewers to upload an unpublished manuscript into a generative AI tool, as is stated in [Elsevier's generative AI policies for journals](#). However, AI tools may be used in a supportive capacity by editors and reviewers to improve structure and language of the review report or editorial decision letter or to conduct a

background literature search. Elsevier is actively evaluating compliant AI tools and may revise this policy in the future.

Preprints

Preprint sharing

Authors may share preprints in line with Elsevier's [article sharing policy](#). Sharing preprints, such as on a preprint server, will not count as prior publication.

We advise you to read our policy on [multiple, redundant or concurrent publication](#).

Use of inclusive language

Inclusive language acknowledges diversity, conveys respect to all people, is sensitive to differences, and promotes equal opportunities. Authors should ensure their work uses inclusive language throughout and contains nothing which might imply one individual is superior to another on the grounds of:

- age
- gender
- race
- ethnicity
- culture
- sexual orientation
- disability or health condition

We recommend avoiding the use of descriptors about personal attributes unless they are relevant and valid. Write for gender neutrality with the use of plural nouns ("clinicians, patients/clients") as default. Wherever possible, avoid using "he, she," or "he/she."

No assumptions should be made about the beliefs of readers and writing should be free from bias, stereotypes, slang, reference to dominant culture and/or cultural assumptions.

These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

Reporting sex- and gender-based analyses

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

- Sex and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals or eukaryotic cells. This should be done in accordance with any requirements set by

funders or sponsors and best practices within a field.

- Sex and/or gender dimensions of the research should be addressed within the article or declared as a limitation to the generalizability of the research.
- Definitions of sex and/or gender applied should be explicitly stated to enhance the precision, rigor and reproducibility of the research and to avoid ambiguity or conflation of terms and the constructs to which they refer.

We advise you to read the [Sex and Gender Equity in Research \(SAGER\) guidelines](#) and the [SAGER checklist](#) (PDF) on the EASE website, which offer systematic approaches to the use of sex and gender information in study design, data analysis, outcome reporting and research interpretation.

For further information we suggest reading the rationale behind and recommended [use of the SAGER guidelines](#).

Definitions of sex and/or gender

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth") and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).
- Gender generally refers to socially constructed roles, behaviors and identities of women, men and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact and how power is distributed in society.

Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

Jurisdictional claims

Elsevier respects the decisions taken by its authors as to how they choose to designate territories and identify their affiliations in their published content. Elsevier's policy is to take a neutral position with respect to territorial disputes or jurisdictional claims, including, but not limited to, maps and institutional affiliations. For journals that Elsevier publishes on behalf of a third party owner, the owner may set its own policy on these issues.

- Maps: Readers should be able to locate any study areas shown within maps using common mapping platforms. Maps should only show the area actually studied and authors should not include a location map which displays a larger area than the bounding box of the study area. Authors should add a note clearly stating that "*map lines delineate study areas and do not necessarily depict accepted national boundaries*". During the review process, Elsevier's editors may request authors to change maps if these guidelines are not followed.

- Institutional affiliations: Authors should use either the full, standard title of their institution or the standard abbreviation of the institutional name so that the institutional name can be independently verified for research integrity purposes.

Studies in humans

Authors must follow [ethical guidelines](#) for research involving humans, their data, and/or their biological materials (e.g. organs, tissues). Authors should ensure that they adhere to the following:

- All studies must be performed in compliance with relevant laws, regulations, and institutional guidelines and that the appropriate institutional committee(s) has/have approved them. For instance, medical research involving humans should be carried out in accordance with the World Medical Association [Declaration of Helsinki: Ethical principles for medical research involving human participants](#) and should follow the International Committee of Medical Journal Editors (ICMJE) [recommendations for the conduct, reporting, editing and publication of scholarly work in medical journals](#) and aim to be representative of human populations in terms of sex, age and ethnicity. Sex and gender terms should be used correctly, as outlined in the “Reporting Sex- and Gender-based analyses” section of this journal’s Guide for Authors.
- Informed consent must be obtained for research with human participants, including their data or biological materials. Consent must be given without coercion, and the privacy rights of human participants must always be observed. If your study contains human participants, please read Elsevier’s full [human research policy](#) for further information.
 - Authors should be aware that written (or electronic) consent is required for certain studies under national laws. In studies where only verbal informed consent was obtained, authors should include justification for the absence of written consent, the name of the approving ethics committee, and a statement on how verbal consent was recorded.
 - Consent for publication should also be obtained.
- Authors must obtain and retain explicit written consent for any identifiable participant materials (e.g. case details, videos, recordings, images, photographs, and illustrations). All participant information must be de-identified: do not include names or personal identifiers (e.g., initials, ID numbers, DOB)
- Images should be used only when scientifically necessary and with explicit consent, and nonessential identifying details must be removed. If identifiers are altered to ensure anonymity, authors must confirm that scientific meaning is not affected. Fully anonymized images (e.g., x-rays, scans) that contain no identifying marks and are not accompanied by identifying text do not require formal consent, though informing the patient is good practice. Case reports always require consent. Simply obscuring identity (e.g., eye bars or blurring) is not sufficient without consent.
- Written consents must be retained by the authors. They should not be provided to this journal unless this is specifically requested in exceptional circumstances, for example, when a legal issue arises. Only then should you provide copies of the consents, or evidence that all relevant consents were obtained.

- Appropriate consents and safeguards should be in place for research with vulnerable populations, as described in Elsevier's [human research policy](#). In many countries, additional ethical approval may be needed for research with those who lack the capacity to consent (e.g. children or those with mental health or neurological conditions including dementia).
- For the full policy, see "[Consent requirements for case reports, personal information, and images.](#)"

All studies involving human participants require a statement that details any ethical approval(s), guidelines followed, confirmation of informed consent, and any other necessary information in line with Elsevier's [human research policy](#). The statement should contain the date and reference number of any ethical approval(s) obtained

If ethical approval is not required by national laws or where a study has been granted an exemption from an ethics committee, the statement must explain why the study is exempt and confirm that the appropriate protocols for protecting the rights and privacy of all participants were utilized during the execution of the research (e.g. no coercion to participate, full disclosure of study requirements and risks, consent of participants (or their Legal Representatives), no release of participant data without their knowledge, etc.).

Human biological material

For all studies that use human organs or tissues, sufficient evidence must be provided that these were procured in line with [WHO Guiding Principles on Human Cell, Tissue and Organ Transplantation](#). The source of the organs or tissues used in clinical research must be transparent and traceable and must not be sourced from executed prisoners or prisoners of conscience, consistent with recommendations by [Global Rights Compliance on Mitigating Human Rights Risks in Transplantation Medicine](#).

For clinical studies, a statement of informed consent having been obtained from a patient or their legal representative, paired with ethical approval for the study from a suitable institution, as required by the policies of the journal, may be considered sufficient evidence, but the journal reserves the right to request additional evidence in cases where it feels this is not sufficient.

If the manuscript describes organ transplantation authors must additionally declare within the manuscript that:

- Autonomous consent, free from coercion, was obtained from the donor(s) or their next of kin.
- Organs and/or tissues were not sourced from executed prisoners or prisoners of conscience.

Studies in animals

All animal experiments should comply with the [ARRIVE \(Animal Research: Reporting of In Vivo Experiments\) guidelines](#) (or national equivalent) and should be carried out in accordance with relevant laws and associated guidelines for animal welfare. This may include [the U.K. Animals \(Scientific Procedures\) Act 1986](#); [the EU Directive 2010/63/EU on the protection of animals used for scientific purposes](#); [the National Research Council Guide for the Care and Use of Laboratory Animals](#); or those of an equivalent internationally recognized body.

Authors must ensure that the manuscript contains a statement that confirms which guidelines have been followed. This statement should include the ethics committee approval number, when available. If a study was granted exemption or did not require ethics approval, a statement detailing this should be included in the manuscript.

The sex of animals, and where appropriate, the influence (or association) of sex on the results of the study should be indicated.

Where applicable, researchers are encouraged to document informed consent from the client or owner in studies involving client-owned animals and adhere to best practices of veterinary care. For research on threatened and/or endangered species, authors are advised to consult international best practice guidelines such as the [International Union for Conservation of Nature \(IUCN\) Policy Statement on Research Involving Species at Risk of Extinction](#) and [IUCN red list guidelines](#).

Delayed publication

If you need to delay the publication of your article for any reason, please contact the editorial office of the journal or our [Journal Article Publishing Support Center](#) at the earliest possible opportunity. While we will try to ensure that your article is not published before a certain date if you make your request on a timely basis, we cannot guarantee this. Please also note that once the agreed period of delay has expired the article may be published at any time thereafter.

Writing and formatting

File format

We ask you to provide editable source files for your entire submission (including figures, tables and text graphics). Some guidelines:

- Save files in an editable format, using the extension .doc/.docx for Word files and .tex for LaTeX files. A PDF is not an acceptable source file.
- Format Word files in a single-column layout. Double-column formatting is only permitted for LaTeX submissions.
- Remove any strikethrough and underlined text from your manuscript, unless it has scientific significance related to your article.
- Use spell-check and grammar-check functions to avoid errors.

We advise you to read our [Step-by-step guide to publishing with Elsevier](#).

Title page

You are required to include the following details in the title page information:

- Article title. Article titles should be concise and informative. Please avoid abbreviations and formulae, where possible, unless they are established and widely understood, e.g. DNA.
- Author names. Provide the given name(s) and family name(s) of each author. The order of authors should match the order in the submission system. Carefully check that all names are accurately spelled. If needed, you can add your name between parentheses in your own script after the English transliteration.

- **Affiliations.** Add affiliation addresses, referring to where the work was carried out, below the author names. Indicate affiliations using a lower-case superscript letter immediately after the author's name and in front of the corresponding address. Ensure that you provide the full postal address of each affiliation, including the country name and, if available, the email address of each author.
- **Corresponding author.** Clearly indicate who will handle correspondence for your article at all stages of the refereeing and publication process and also post-publication. This responsibility includes answering any future queries about your results, data, methodology and materials. It is important that the email address and contact details of your corresponding author are kept up to date during the submission and publication process.
- **Present/permanent address.** If an author has moved since the work described in your article was carried out, or the author was visiting during that time, a "present address" (or "permanent address") can be indicated by a footnote to the author's name. The address where the author carried out the work must be retained as their main affiliation address. Use superscript Arabic numerals for such footnotes.

Abstract

You are required to provide a concise and factual abstract which does not exceed 250 words. The abstract should briefly state the purpose of your research, principal results and major conclusions. Some guidelines:

- Abstracts must be able to stand alone as abstracts are often presented separately from the article.
- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).
- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

Keywords

You are required to provide 1 to 7 keywords for indexing purposes. Keywords should be written in English. Please try to avoid keywords consisting of multiple words (using "and" or "of").

We recommend that you only use abbreviations in keywords if they are firmly established in the field.

Graphical abstract

You are required to provide a graphical abstract at submission.

The graphical abstract summarizes the contents of your article in a concise, pictorial, professional form, highlighting important aspects of the research. It is designed to appeal to an interdisciplinary audience, and draw more attention to your online article. Some guidelines:

- Ensure you have obtained the necessary permission to include any third party owned material in a graphical abstract.
- Use of generative AI or AI-assisted tools in the production of graphical abstracts must align with Elsevier's [GenAI Policies for Journals](#).

- Submit your graphical abstract as a separate file in the online submission system.
- Ensure the image is 531 x 1328 pixels (h x w) or proportionally more, and is readable at a size of 5 x 13 cm.
- Preferred file types for graphical abstracts are TIFF, EPS, PDF or MS Office files.

More information, examples and a template can be found in the [graphical abstract guidelines](#).

Tables

Tables must be submitted as editable text, not as images. Some guidelines:

- Place tables next to the relevant text or on a separate page(s) at the end of your article.
- Cite all tables in the manuscript text.
- Number tables consecutively according to their appearance in the text.
- Please provide captions along with the tables.
- Place any table notes below the table body.
- Avoid vertical rules and shading within table cells.

We recommend that you use tables sparingly, ensuring that any data presented in tables is not duplicating results described elsewhere in the article.

Figures, images and artwork

Figures, images, diagrams and other graphical media (artwork) must be supplied as separate files along with the manuscript, in line with these [artwork and media instructions](#). Some excerpts:

When submitting artwork:

- Cite all images in the manuscript text.
- Number images according to the order they appear within your article.
- Submit each image as a separate file using a logical naming convention for your files (for example, Figure_1, Figure_2 etc).
- Text graphics may be embedded in the text at the appropriate position. If you are working with LaTeX, text graphics may also be embedded in the file.

Captions

All artwork must have a caption. A caption should consist of a brief title (not displayed on the figure itself) and a description of the image. Keep the amount of text in any image to a minimum, though any symbol or abbreviation

should be explained.

Read how to add captions to your submission [here](#).

Formats

When your artwork is finalized, "save as" or convert your electronic artwork to the formats listed below taking into account the given resolution requirements for line drawings, halftones, and line/halftone combinations:

- Vector drawings: Save as EPS or PDF files embedding the font or saving the text as "graphics."
- Color or grayscale photographs (halftones): Save as TIFF, JPG or PNG files using a minimum of 300 dpi (for single column width: min. 1063 pixels, full page width: 2244 pixels).
- Bitmapped line drawings: Save as TIFF, JPG or PNG files, using a minimum of 1000 dpi (for single column width: min. 3543 pixels, full page width: 7480 pixels).
- Combinations bitmapped line/halftones (color or grayscale): Save as TIFF, JPG or PNG files using a minimum of 500 dpi (for single column: min. 1772 pixels, full page width: 3740 pixels).

Please do not submit:

- Files that are too low in resolution.
- Artwork where text is disproportionately small compared to the image size, as text may become unreadable.
- Different images or graphs combined into one, as this affects accessibility

Color artwork

If you submit usable color figures with your accepted article, we will ensure that they appear in color online.

Please ensure that color images are accessible to all, including those with impaired color vision. Learn more about [color and web accessibility](#).

Generative AI and Figures, images and other artwork

Please read our policy on the use of generative AI and AI-assisted tools in figures, images and artwork, which can be found in Elsevier's [GenAI Policies for Journals](#). This policy states:

- AI tools may be used to support the creation of explanatory images (such as flow charts, conceptual diagrams, or schematic illustrations)
- AI tools may be used to create data visualizations (such as plots, charts, graphs, or heatmaps) when the output is directly derived from underlying data using reproducible analytical, computational, or statistical methods.
- AI tools must not be used to create or alter images that represent primary observed or experimental data (such as western blots, microscopy images, histology images, or patient images) that were not directly obtained in the

research. This includes adjustments to brightness, contrast, or color balance, which should only be done using established image processing software.

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